AIORI-3 HACKATHON / PS-035 / LEVEL 1

Evidence-Based Operator Network Fault Localisation

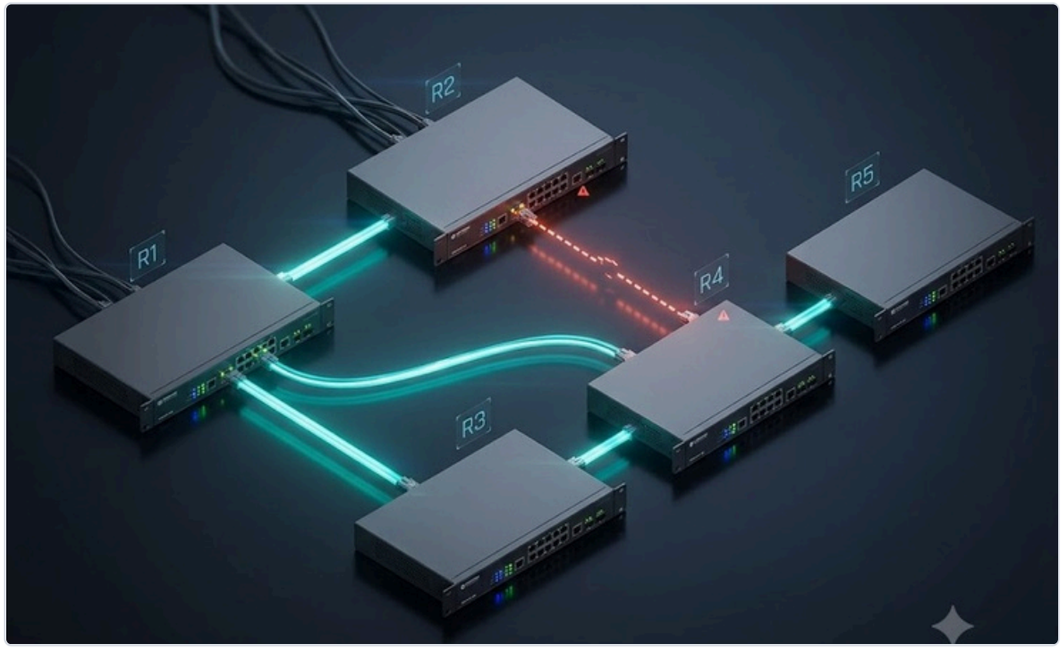
Intelligent Network Fault Diagnosis and Root-Cause Analysis

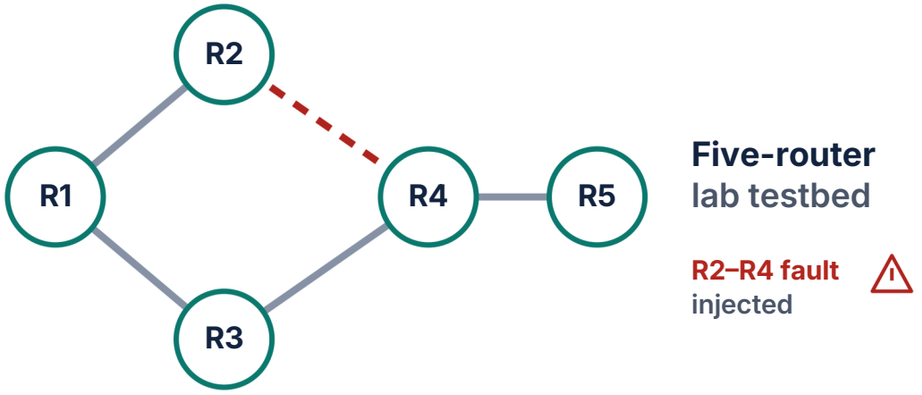
TEAM-ZNMD

Teamcode: A3-PS035-TC262

Institution: IIIT Bhagalpur · Submission date: 14 October 2026

Project link (GitHub)





ABSTRACT

Problem Description

Afive-node fault-localisationsystem thatcombines **topology, syslog, BGP and ping** evidence, ranks suspected causes and explains each assessment.

One fault, many alarms	One link failure triggers several downstream alarms.
Alerts are not causes	Isolated alerts do not establish the root cause.
Evidence can fail	Missing or conflicting evidence requires abstention.

Lab configured; collection, diagnosis, dashboard planned. Example illustrative. [1, 5]



ONE FAULT → MANY ALARMS

ROOT CAUSE
R2–R4link / interface failure

▼

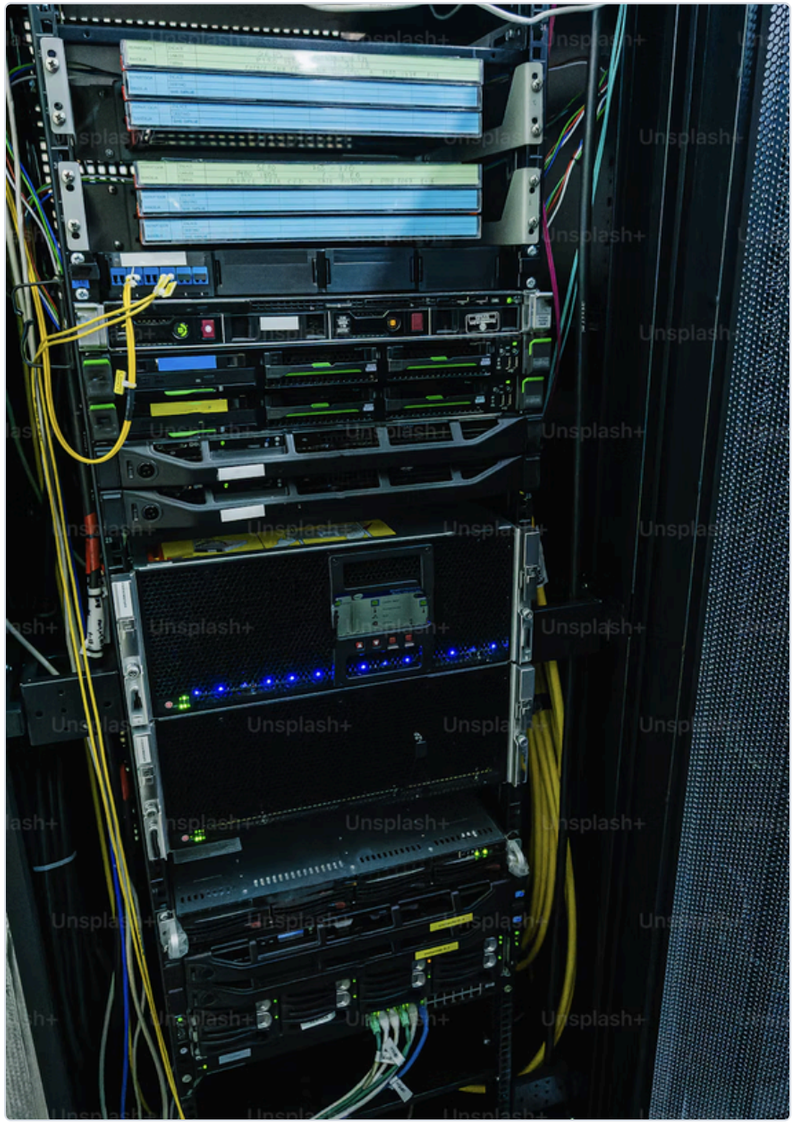
SYMPTOM
Interfacedown on both endpoints

▼

SYMPTOM
BGPsession lost · route change

▼

SYMPTOM
Pingloss · latency alarms



Background

Syslog events
Timestamped router messages: interface up/down, BGP neighbor changes.

Interface health
Operationalstate,error counters and packet drops.

BGP sessions
Neighbor state,reachability and next-hop changes. [1, 4]

Active probes
Pingreachabilityand latency between nodes.

Topology correlation
Link symptoms tosharednodes and links: one fault, many alarms.

Detect, then diagnose
Detection flagsachange.Diagnosis ranks causes, or abstains.

A failed probe is a symptom. Its cause needs corroborating evidence.

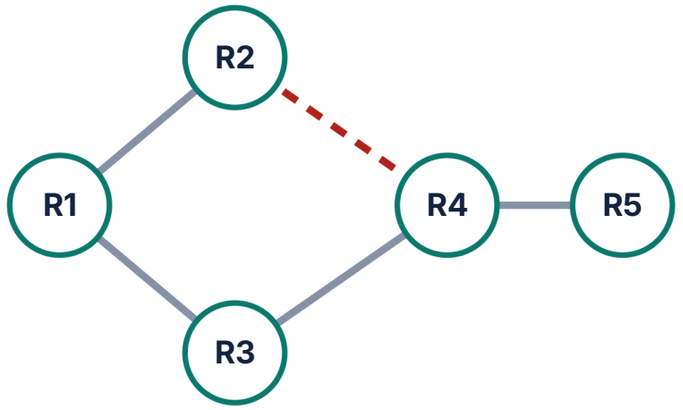
Literature & RFC Review

Reference	Relevant concept	Implication for our design
[1] RFC 4271	BGP state machine and route selection	Read peer state with route evidence
[2] RFC 8632	Alarm identity, severity and lifecycle	Keep resource, type, time and clear state
[3] Containerlab docs	Declarative nodes and links	Rebuild the same test topology
[4] FRRouting docs	BGP operational commands / JSON	Parse structured evidence where supported

DESIGN BOUNDARY

Standards guide our evidence model. RFC 8632 implementation or conformance is not claimed.

Proposed Solution



Five-router
lab testbed

R2–R4 fault
injected

1. Record a healthy baseline
2. Inject one controlled fault
3. Collect logs, state and probes
4. Correlate symptoms in time + topology
5. Rank causes with transparent rules
6. Show evidence, or report insufficient evidence

1 · R2–R4 link / interface fault
Endpoint state + peer event agree

2 · BGP session fault
Weaker: interface evidence also changed

3 · R2 node outage
Contradicted: R2 remains reachable

Boundary
Link vs interface stays ambiguous. No auto-remediation. Illustrative. [5]

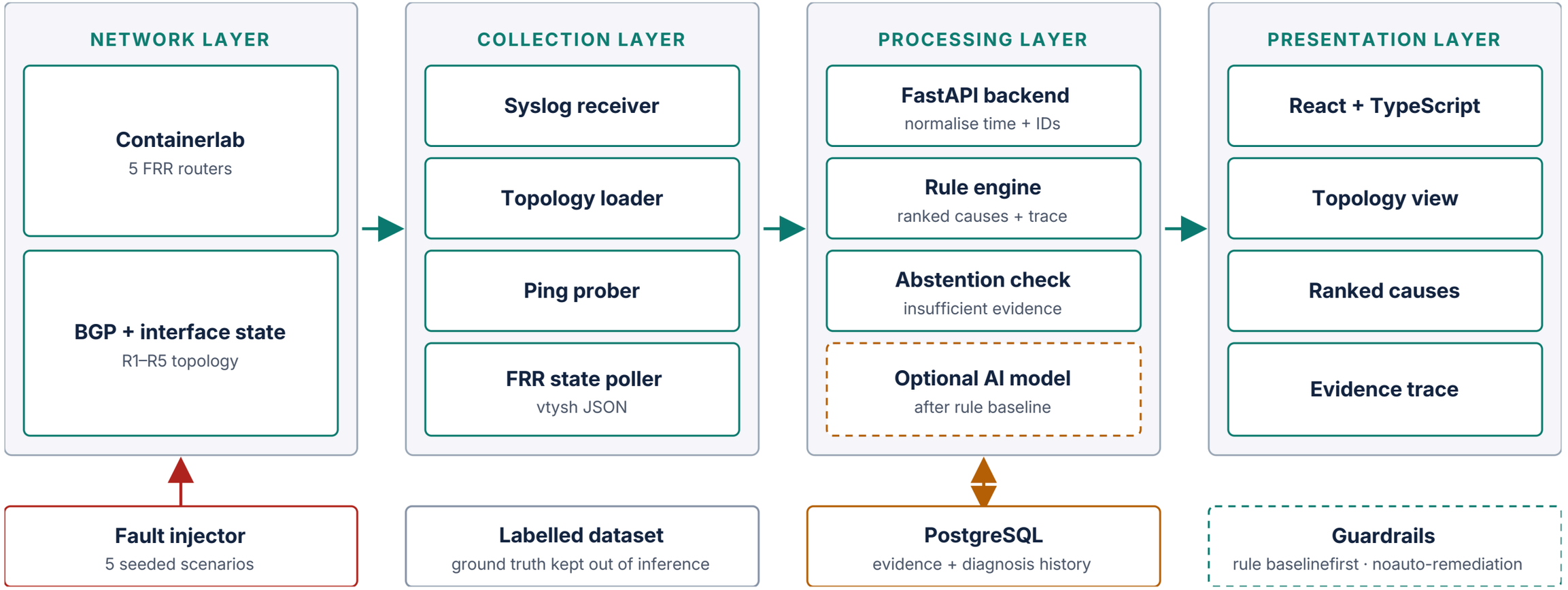
Optimization Proposed

Baseline=ourmanualworkflow,notall monitoringproducts.Benefitsremain expected until measured.

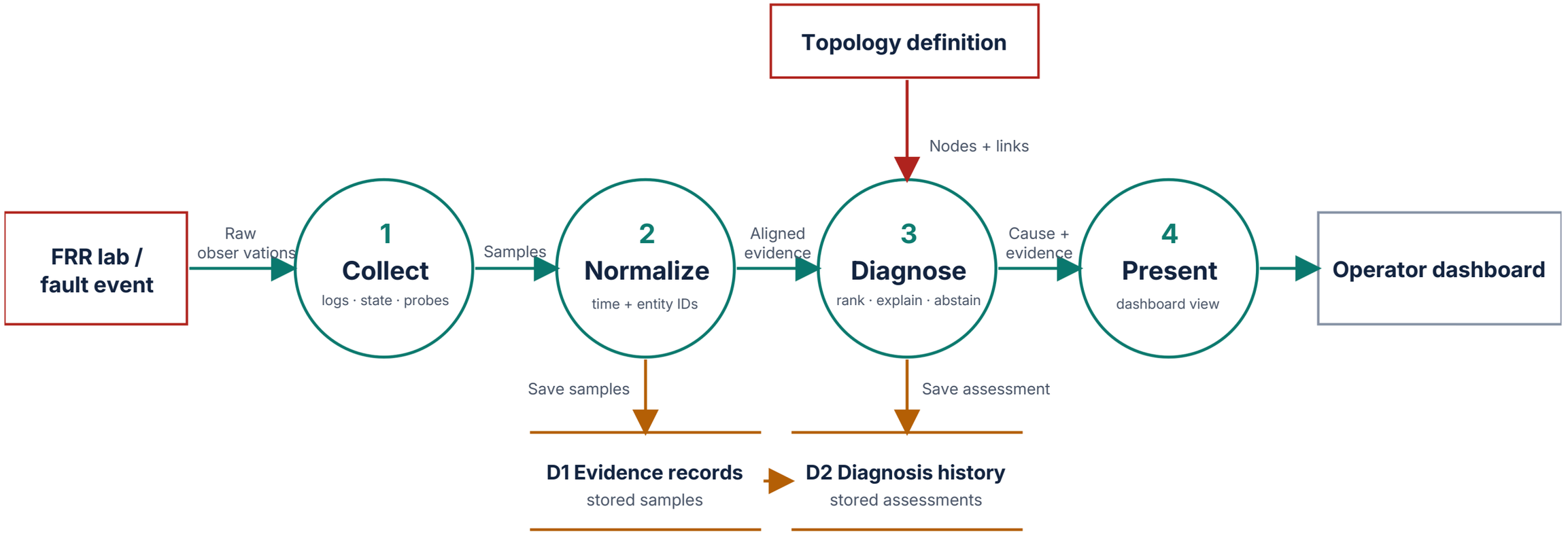
Dimension	Baseline (manual)	Proposed	Expected benefit	Validation
Collection	Ad hoc CLI checks	→ **Timestamped snapshots**	More complete evidence	**Missing-sample rate**
Correlation	Manual comparison	→ **Time + topology grouping**	Less duplicate triage	**Grouped vs raw alerts**
Localisation	Operator hypothesis	→ **Ranked rule matches**	Consistent decisions	**Top-1 / top-3 accuracy**
Explainability	Free-text notes	→ **Evidence + rule trace**	Auditable reasoning	**Trace completeness**
History	Separate log files	→ **PostgreSQL records**	Replayable analysis	**Replay consistency**

Held-out top-1 / top-3 accuracy | Abstention rate | Detection / diagnosis delay | False-positive rate

Solution Architecture & Design



Data Flow Diagram



Rectangle = external entity · Circle = process · Open bar = data store (logical, in PostgreSQL)

Ground-truth labels stay outside inference

Deployment Timeline

01
Network lab

COMPLETED

Topology + FRR configs in GitHub; connectivity and BGP reported

02
Telemetry collector
PLANNED

Syslog + topology + ping snapshots; timestamps and missing-data flags

03
Fault injection

PLANNED

Five seeded faults with recovery scripts; labelled dataset

04
Diagnosis engine
PLANNED

Rules + ranked candidates + abstention; unseen fault placements

05
Dashboard + API
PLANNED

Topology, alerts, evidence trace; replay a stored incident

06
Evaluation

PLANNED

Held-out top-1 / top-3; report + demo + pseudocode [5]

Current State

01 | Network Lab
COMPLETED

02 | Telemetry Collector
PLANNED

03 | Fault Injection
PLANNED

04 | Diagnosis Engine
PLANNED

05 | Dashboard + API
PLANNED

06 | Evaluation
PLANNED

Bibliography

[1] Rekhter, Li & Hares (eds.), Jan 2006

RFC 4271: A Border Gateway Protocol 4 (BGP-4)
rfc-editor.org/rfc/rfc4271

[2] Vallin & Bjorklund, Sep 2019

RFC 8632: A YANG Data Model for Alarm Management
rfc-editor.org/rfc/rfc8632

[3] Containerlab documentation

Topology definition
containerlab.dev/manual/topo-def-file/

[4] FRRouting documentation

BGP; verify against deployed version
docs.fr routing.org/en/latest/bgp.html

[5] AIORI-3 problem statement

PS-035:Evidence-BasedOperatorNetwork Fault Localisation

Access note

Web referencesaccessed 13 October 2026. Implementation status follows the supplied project brief.

TEAM-ZNMD

Thank You for Your Attention!

Towards faster, evidence-driven network fault diagnosis.

Intelligent Network Fault Diagnosis and Root-Cause Analysis

